



CAPSTONE PROJECT DOCUMENTATION

LifeRise Initiative: A Blockchain-Based Economic Empowerment
Ecosystem for Stroke Survivors in Zambia

Prepared by
Michelo Saviour Hanamwinga
Collegiate of Science in Blockchain Studies

Professor: Prof. Johannes Dowe
Capstone Date: 29 July 2026

Technical Documentation • Architecture • Security • Data Management • Smart Contracts • Token Utility

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3. Abstract

LifeRise Initiative is a proposed digital ecosystem intended to support the economic empowerment of stroke survivors in Zambia. The project uses blockchain selectively, rather than treating blockchain as a replacement for a normal database or backend. The architecture is hybrid: a mobile application provides the user interface; the backend controls application workflows and authorization; PostgreSQL stores private and frequently changing information; and an EVM-compatible blockchain provides a public, tamper-resistant record for selected transactions and utility-token activity.

The most important engineering principle is separation of concerns. Personal information remains off-chain. Blockchain is used only where an independently verifiable record, deterministic execution or token accounting creates a meaningful benefit. The LifeRise Token (LRT) is a utility token intended to support ecosystem participation and rewards. It is not designed to be digital money, a stablecoin, or a 1:1 representation of the Zambian Kwacha, US Dollar or another currency.

This documentation expands on the presentation slides and records the reasoning, technical workflow, security assumptions, data boundaries and blockchain design decisions behind the LifeRise platform.

4. Capstone Defense Approval Form

The Capstone Defense Approval Form is obtained from the Learning Management System (LMS). It is completed and signed by the appropriate academic authorities prior to the defense and forms part of the official submission package.

CAPSTONE DEFENSE APPROVAL FORM

Name:

Capstone Defense Approval Date & Time:

Degree: Nanodegree **Collegiate of Science**

Unit: College of Continuing and Professional Studies (CAPS)

Capstone Title:

Abstract:

In partial fulfillment of the requirements for the nanodegree of

(program)

We, the undersigned, recommend that the capstone project completed by the student listed above be acknowledged and counted as a requirement for graduation.

COMMITTEE APPROVAL

adviser name

signature

member

signature

member

signature

*At least one of the signatures above must be that of a member of the Althash University Faculty. By completing this box, you are confirming that the student above has satisfactorily completed the academic work for the program stated above. The submission of this form indicates the approval of the format and content of this document. This form is required for completion of the capstone project deposit.

5. CODE 499 Waiver Request Form

The CODE 499 Waiver Request Form is obtained from the LMS. It documents any approved fee waiver related to the capstone requirements and forms part of the official submission package.



CODE 499 WAIVER REQUEST FORM	
Name:	
Capstone Defense Approval Date & Time:	
Degree: Nanodegree	
Unit: College of Continuing and Professional Studies (CAPS)	
Capstone Title:	
Abstract:	
<i>I, the undersigned, request that the student indicated above be granted a fee waiver for the capstone project he or she completed on time and in accordance with cohort standards.</i>	
APPROVAL	

adviser name

signature

6. Introduction

The LifeRise Initiative addresses a social-impact problem through a technology architecture that combines established software engineering practices with blockchain concepts studied during the Blockchain Studies program.

Blockchain is often presented as a solution to every data problem. LifeRise instead treats blockchain as one component in a larger system. This distinction demonstrates practical understanding: a blockchain has strengths—integrity, shared verification and deterministic contract execution—but it also has limitations such as public visibility, gas costs, latency, and inability to directly query ordinary databases.

Project Objective

The objective is to design a platform that can support transparent participation, selected rewards and accountable transactions while protecting private user information and maintaining a realistic technical architecture.

7. Problem Statement

Stroke survivors can face reduced employment opportunities, financial dependency, limited access to support and difficulty participating in sustainable economic activities. Support programs may also need better mechanisms for tracking participation, documenting transactions and demonstrating accountability.

The problem is therefore not simply a lack of technology. It is the need for a trustworthy system that can combine privacy-sensitive application data with transparent and verifiable records where transparency is useful.

8. Methodology & Solution to the Problem

LifeRise proposes a platform through which beneficiaries and other ecosystem participants can register, participate in approved programs, receive support or rewards, and interact with participating organizations.

The solution uses three complementary mechanisms:

- **Conventional application infrastructure** for accounts, user interfaces, business workflows and private data.
- **Blockchain infrastructure** for selected verifiable transactions and token state.
- **Smart contracts** for deterministic rules that genuinely benefit from on-chain execution.

The design intentionally avoids putting medical or personally identifying information on a public blockchain. The architecture is hybrid because a fully on-chain design would expose sensitive information and increase costs, while a fully off-chain design would lose the transparency and independent verification benefits that blockchain can provide.

9. Vision, Mission, Goals, and Objectives

Vision

A Zambia in which stroke survivors regain economic independence, participate meaningfully in community life, and access transparent, accountable support systems enabled by appropriate technology.

Mission

To design and deliver a hybrid blockchain-enabled platform that empowers stroke survivors through verified participation, programmable incentives, privacy-preserving data management, and accountable transaction records.

Goals

- Enable secure registration and role-based participation for beneficiaries, volunteers, donors and administrators.
- Provide transparent, auditable reward and participation mechanisms via a utility token (LRT).
- Keep all personal and medical information off-chain while using blockchain only for verifiable state.
- Demonstrate a technically defensible, security-conscious architecture suitable for real-world pilot deployment.

Objectives

- Complete a working hybrid architecture (mobile UI, backend, PostgreSQL, EVM smart contracts).
- Implement authentication, authorization and wallet-based signing without exposing private keys.
- Define and document token utility, supply policy and allocation categories prior to deployment.
- Conduct phased testing and a limited pilot before any wider public launch.

10. Token Name

LifeRise Utility Token (also referred to as the LifeRise Token).

Classification: **Utility Token**. It is intended to have utility inside the LifeRise ecosystem and to create a programmable incentive and participation mechanism rather than a parallel monetary system.

11. Token Ticker

LRT

The ticker LRT is used consistently across documentation, smart-contract interfaces and application displays.

12. Token Maximum Supply

Maximum supply is a design decision derived from expected participation, reward frequency, program duration and sustainability. It is not selected merely because a large number sounds attractive.

If LifeRise adopts a fixed maximum supply, the smart contract will enforce a hard cap. A mint function must reject any operation that would cause total supply to exceed that cap. If there is no minting after deployment, the deployment and allocation policy will state this clearly.

Defense position: The exact numerical cap is a tokenomics parameter under final design. It is preferable to state that the cap remains under design justification rather than present an arbitrary number as finalized.

A final token allocation will be documented before deployment. Categories may include ecosystem rewards, community incentives, partnerships and development reserves, each with a defined purpose and, where appropriate, release or vesting rules.

13. Project Budget and Budget Allocation

For the pilot concept, the working budget is **K36,000** (Zambian Kwacha), corresponding to approximately **US\$1,800** at a working conversion rate of K20 per US\$1.

Category	Amount (ZMW)	Amount (USD)	Purpose
Survivor payments / support	K27,000	US\$1,350	Direct support and rewards for beneficiaries
Coordination	K6,000	US\$300	Project coordination and operational oversight
Communication	K3,000	US\$150	Outreach, materials and stakeholder communication
Total	K36,000	US\$1,800	Pilot working budget

Important: These figures should be replaced if the final approved budget differs. The final submission uses one consistent set of figures across slides and this PDF.

14. Token Slogan & Description

Slogan / Tagline

“Empowering Stroke Survivors. Restoring Independence. Creating Opportunity.”

Description

LRT is a utility token designed to reward verified participation in approved LifeRise activities, recognize contributions by volunteers or community participants, provide access to selected platform features or participation tiers, support transparent and auditable incentive distribution, and create a measurable participation record associated with an ecosystem wallet.

What LRT is NOT:

- It is not digital money for general everyday payments.
- It is not a stablecoin.
- It is not backed or pegged 1:1 to ZMW, USD or another fiat currency.
- It should not be described as a guaranteed monetary store of value.

15. Project Launch Date

The project follows a controlled, phased implementation rather than an immediate full public deployment.

- **Phase 1** – Requirements and research
- **Phase 2** – Architecture and design
- **Phase 3** – Development (mobile interface, backend, database, smart-contract prototype)
- **Phase 4** – Testing (authentication, authorization, contract functions, edge cases)
- **Phase 5** – Pilot (limited group, feedback, usability and operational measurement)
- **Phase 6** – Evaluation and scaling

Capstone presentation and documentation date: **29 July 2026**. A specific public pilot launch date will be set after successful completion of testing and internal evaluation.

16. Token Logo

The official LifeRise Initiative visual identity features a stylized figure rising from an open hand within a circular emblem, rendered in purple and green. The wordmark “LifeRise” appears with the subtitle “INITIATIVE ZAMBIA” and the tagline “Empowering Stroke Survivors. Restoring Independence. Creating Opportunity.”



High-resolution logo files are maintained in the project repository and branding package for consistent use across the mobile application, documentation and presentation materials.

17. GitHub Account & Link

GitHub provides version control, project organization and evidence of development. The repository contains a clear README describing the project, its purpose, architecture and current status. A logical structure separates documentation, smart contracts, backend services and mobile application code. Commit history shows meaningful development progress.

GitHub Repository:

<https://github.com/mitchello2022-png/LifeRise-initiative-.git>

For the capstone defense, the repository link and a screenshot of the repository home page provide evidence that the project artifacts are organized and accessible.

18. List of Courses Taken

The academic component of the capstone is supported by completed Blockchain Studies coursework. Relevant topics include:

- Blockchain foundations
- Cryptography, public/private keys and digital signatures
- Hash functions
- Cryptocurrency and tokenomics
- Consensus mechanisms
- Wallet safety
- Interoperability and scalability
- Decentralized models and enterprise blockchain
- Blockchain architecture
- Smart contracts, Ethereum, Hyperledger Fabric
- Self-sovereign identity
- Blockchain-enabled electronic health records
- NFT development
- Blockchain leadership and management
- Blockchain profession and career development
- Directed research and capstone defense

B-Individual Requirements

19. Microcredentials & Link

Supporting microcredentials obtained during the Blockchain Studies program are included in the documentation package. Links and verification details are provided with the official credential documents.

[Attach microcredential certificates and verification links.]

20. Ladderized Certificates, Diploma & Link

The documentation package includes the Certificate of Satisfactory Competence in Blockchain Studies, Certificate of Outstanding Competence in Blockchain Studies, and the Collegiate of Science in Blockchain Studies Diploma, together with verification links where available.

[Attach certificates, diploma and verification links.]

21. Transcripts of Academic Records

The official transcript of academic records is included in the final submission package where available.

[Attach official transcript.]

22. BCE 501 Blockchain Continuing Education™

Evidence of completion or enrollment in BCE 501 Blockchain Continuing Education™ is documented as part of the academic record supporting the capstone.

[Attach BCE 501 evidence / certificate.]

23. Professional Organization (Optional)

Membership in professional blockchain or technology organizations is optional. Any relevant memberships or affiliations may be listed here.

[Optional – list professional organizations if applicable.]

24. Resume & Link

A current professional resume / curriculum vitae is included in the submission package. An online version or portfolio link may be provided.

[Attach resume and any public link.]

25. Blockchain Oath of Profession

The Blockchain Oath of Profession affirms commitment to ethical practice, transparency, security and the responsible application of blockchain technology. The signed oath forms part of the capstone package.

[Attach signed Blockchain Oath of Profession.]

26. Capstone Defense Awareness Plan

The Capstone Defense Awareness Plan outlines preparation steps, key talking points, anticipated questions and demonstration readiness for the formal defense. It ensures the candidate can clearly articulate architecture decisions, security rationale, token design and the hybrid on-chain / off-chain boundary.

[Attach Capstone Defense Awareness Plan document.]

Platform and Technical Overview

The platform is understood as a set of cooperating layers. Each layer has a defined responsibility and a defined boundary. A secure system should not depend on one technology doing everything.

Layer	Primary responsibility	Should NOT be responsible for
Mobile app	User interaction, display, request initiation	Final authorization or secret-key custody
Backend	Business logic, API, authorization, database access	Authoritative blockchain state without verification
PostgreSQL	Private application data and workflow records	Public immutable transaction proof
Blockchain	Verifiable on-chain state and transaction history	Private profiles or medical records
Smart contract	Token accounting and predefined on-chain rules	Direct PostgreSQL queries, private data, full app logic

Detailed End-to-End System Workflow

Step 1 – Registration

User creates an application account. Backend validates data and creates an internal user record. Passwords are hashed with a strong algorithm; never stored as plain text.

Step 2 – Authentication

User proves knowledge of credential. Server verifies password hash and establishes an authenticated session or token. MFA can be applied for higher-risk accounts.

Step 3 – Application request

Authenticated user performs an action (e.g., enrolling or requesting a reward). Mobile app sends the request to the backend over HTTPS/TLS.

Step 4 – Backend validation

Backend checks permission, request validity, program existence and whether the action has already been processed. Client-side validation alone is not a security boundary.

Step 5 – Database lookup

Backend retrieves required records from PostgreSQL (internal identifier, role, program participation, reward eligibility). Sensitive information stays inside the controlled environment.

Step 6 – Blockchain decision

If the action requires a verifiable on-chain record or token operation, the backend prepares the blockchain interaction. Otherwise the action remains entirely off-chain.

Step 7 – Wallet authorization

Where the user must authorize a blockchain action, the user's wallet signs the transaction. The application never asks the user to reveal the private key.

Step 8 – Smart-contract execution

The network executes the smart-contract function. The contract checks its own on-chain rules (access control, balances, limits, maximum-supply constraint).

Step 9 – Confirmation and event

After inclusion and required confirmations, the application associates the transaction hash and relevant contract event with the user's off-chain record.

Step 10 – User feedback

The application displays transaction status or reward. The user can independently inspect the public record while the private database record remains protected.

Important boundary: *The smart contract does not reach backward into PostgreSQL to look up a user's medical record. If off-chain information must affect an on-chain decision, the system needs an explicit trusted mechanism to bring an appropriate result on-chain.*

Authentication and Authorization

Application authentication answers: “Who is this user?” Credentials use strong password hashing and secure session management. Two-factor authentication can be applied to privileged accounts.

Authorization answers: “What is this authenticated user allowed to do?” Role-based access control distinguishes beneficiaries, donors, volunteers, administrators and other roles. Authorization is enforced on the server, not merely hidden in the UI.

A blockchain wallet is controlled by a cryptographic private key. A digital signature proves that the holder of the corresponding private key authorized a message or transaction. The platform never collects or stores a user's private key.

Security does not depend on keeping the architecture secret. The design remains secure even if an attacker knows the API routes, database technology, contract address and general workflow.

Data Management and Database Responsibilities

PostgreSQL (off-chain): Private, relational, frequently updated data — account details, contact information, roles, program records, internal identifiers, support records.

On-chain: Minimum state required for decentralized function — wallet addresses, token balances, reward transactions, timestamps and contract events.

The backend maintains an internal relationship between an application record and a blockchain transaction hash, allowing statements such as “reward record X corresponds to transaction Y” without publishing the private profile.

Blockchain Layer and Transaction Lifecycle

A blockchain transaction is signed, broadcast, validated by consensus, included in a block and confirmed according to network rules. A transaction can fail even though correctly signed (e.g., unmet require conditions, insufficient balance, out of gas). The backend therefore treats blockchain confirmation and transaction success as separate states.

Smart Contract Design, Necessity and Limitations

The contract is justified when LifeRise needs a shared, independently verifiable rule set (token accounting, predefined reward rules).

What the contract should do: Maintain token balances; apply predefined reward rules; enforce access control on privileged functions; emit events; enforce maximum supply if a cap is used.

What the contract should not do: Store medical records or personal profiles; query PostgreSQL; replace the entire backend; assume nobody can call its public functions; perform unnecessary on-chain calculations.

Security Model and Attack-Vector Analysis

The security model assumes that attackers may know the architecture. Security therefore comes from cryptographic protections, access controls, secure implementation and operational controls—not from hiding the system design.

Attack vector	What an attacker tries	Primary control
Credential attack	Steal/reuse passwords	Strong hashing, MFA, rate limiting, secure sessions
SQL injection	Manipulate database queries	Parameterized queries / ORM, validation
Broken authorization	Access another user's records	Server-side RBAC and object-level checks
API abuse	Replay or automate requests	Rate limiting, validation, idempotency, monitoring
Private-key theft	Authorize blockchain actions	Never collect keys; secure wallet/key custody
Unauthorized contract call	Invoke privileged function	On-chain access-control checks
Duplicate reward	Claim same reward repeatedly	Unique activity IDs, state checks, nonces/idempotency
Contract bug	Exploit token/reward logic	Minimal code, tests, review/audit before deployment
Privacy leakage	Read sensitive public data	Keep sensitive data off-chain; minimize on-chain metadata

Conclusion

LifeRise demonstrates a practical application of blockchain rather than a blockchain-for-everything approach. The project separates private application data from public, verifiable blockchain state.

The mobile application and backend remain responsible for normal application operations. PostgreSQL handles private structured data. Blockchain provides a tamper-resistant transaction layer, while smart contracts automate only the rules that genuinely benefit from on-chain execution.

LRT is positioned as a utility token that supports participation and rewards. It is deliberately not a stablecoin, not a 1:1 representation of fiat currency and not the sole foundation of the platform. The resulting architecture is intended to be secure, privacy-aware, economically conscious and technically defensible.

Prepared for Capstone Defense • LifeRise Initiative Zambia

Michelo Saviour Hanamwinga • Collegiate of Science in Blockchain Studies

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